



Okan Berhoğlu

Robotics Engineer

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<https://okanberhoglu.github.io/>

PROFILE

Robotics Engineer with experience in robotics research, industrial software development, system identification, and autonomous systems. Skilled in robot software, industrial communication, data acquisition, simulation, and system integration, with hands-on experience developing and deploying engineering software. Background in Mechanical Engineering with an M.S. in Robotics.

EDUCATION

Middle East Technical University *Ankara, Turkey*
Master of Science, Robotics
2024 – 2026

TOBB Economics and Technology University
Ankara, Turkey
Bachelor of Science, Mechanical Engineering
2018 – 2023

Atatürk University *Remote*
Associate Degree, Computer Programming,
Second University Program
2021 – 2023

SKILLS

Software: Java, C#, Python, C++, REST API, Git, Linux, CUDA, Unity, ROS/ROS2, MATLAB, Simulink, Simscape, SQL, InfluxDB

Industrial Communication: TCP/IP, UDP, EtherNet/IP, Modbus TCP, CAN, RS-485

Language: English - Upper-Intermediate, TOEFL iBT 2026 score: 5/6 (equivalent to 100/120 in old score system)

PROJECTS

Development of a Gimbal and Its Digital Twin
Designed and developed a gimbal and its digital twin enabling remote control of a physical gimbal's direction, featuring a React.js web application with a 3D model (built in Three.js) that mirrors the gimbal's movements in real-time.

Customizable Laptop Stand MATLAB App
Developed a user-friendly MATLAB app that generates a customizable 3D laptop stand model in STL format based on user-defined dimensions, using a 3D topology optimization script for the mechanical design and parametrically modeling the resulting shape with Bezier surfaces.

ChatMoonVLM

Developed a locally-run visual question answering web app that uses a vision-language model to let users upload an image and hold a conversation.

EXPERIENCE

Middle East Technical University *Ankara, Turkey*

Researcher *October 2025 - Present*

- Conduct research on mobile robotic systems, including kinematic and dynamic modeling, simulation, sensor integration, and autonomous control algorithm development.
- Contribute to a research project investigating human rhythmic behavior. Within the project, I developed an experimental setup using a 3-axis haptic device, collected data from predefined rhythmic motions, and conducted system identification and behavior modeling studies.

Teknopar Industrial Automation *Ankara, Turkey*

Software Engineer *February 2024 – October 2025, 1 year 9 months*

- Designed and developed a Windows service-based data acquisition application for CNC controllers. Integrated TCP/UDP-based industrial communication protocols to collect real-time machine data, standardized the collected data into an MTConnect-like format, and actively contributed to the software's licensing, deployment, and commissioning processes.
- Developed an OSGi-based application with a Swing user interface for military truck systems, enabling operators to monitor onboard power units and control the vehicle's electrical system. Integrated industrial communication protocols such as EtherNet/IP and Modbus TCP for device communication and data collection.
- Actively contributed to the proposal writing of a TÜBİTAK-funded project focused on developing the system model, autonomy stack, and digital twin of a mobile robot. Within the project, worked on AI-based environmental perception and autonomous navigation using ROS2 as a software framework, 3D LiDAR, and depth camera sensors.

Candidate Engineer *December 2022 – February 2024, 1 year 3 months*

- Supported R&D activities focused on collecting data from CNC machines and industrial systems.
- Contributed to the modeling of an electromechanical system using Simulink and Simscape, and developed its integration into IBM Rhapsody.
- Analyzed experimental data from hardware prototypes and supported grey-box system identification and parameter estimation studies.

Intern *May 2022 – December 2022, 8 months*

- Worked on data acquisition methods from CNC machines and supported early industrial machine communication studies.
- Developed a MATLAB Simscape Multibody model of an electromechanical system, integrating CAD geometry and dynamic parameters to simulate and analyze real-world motion.
- Contributed to mechanical design by creating hydraulic system components in SolidWorks.

Anova R&D Technologies *Ankara, Turkey*

Intern *May 2021 – August 2021, 4 months*

- Gained hands-on experience in machining, quality control, and finishing processes, and contributed to mold and fixture design activities.